

PHYSM0071: Coursework 1
Correlations and fluctuations in a two-dimensional lattice gas
Model report for marking (base seed 2468135)

Mapping and predictions

Let $V = L^2$, let the square-lattice coordination number be $q = 4$, and write $c_i = (1 + s_i)/2$. Since there are $qV/2$ bonds,

$$\sum_{\langle ij \rangle} c_i c_j = \frac{1}{4} \sum_{\langle ij \rangle} s_i s_j + \frac{q}{4} \sum_i s_i + \frac{qV}{8}, \quad N = \frac{V}{2} + \frac{1}{2} \sum_i s_i.$$

The grand-canonical Hamiltonian therefore becomes

$$E - \mu N = -\frac{\epsilon}{4} \sum_{\langle ij \rangle} s_i s_j - \left(\frac{q\epsilon}{4} + \frac{\mu}{2} \right) \sum_i s_i - \frac{qV\epsilon}{8} - \frac{\mu V}{2}.$$

Comparison with $\mathcal{H}_I = -J \sum_{\langle ij \rangle} s_i s_j - H \sum_i s_i$ gives $J = \epsilon/4$, $H = q\epsilon/4 + \mu/2$, and the configuration-independent term $C = -qV\epsilon/8 - \mu V/2$. The factor $\exp(-\beta C)$ cancels from normalised probabilities, so C does not affect configurational averages or phase behaviour, although it is required when relating the two partition functions. Coexistence corresponds to Ising zero field, so $\mu_{\text{coex}} = -q\epsilon/2 = -2\epsilon$. Also

$$\rho = \frac{1+m}{2}, \quad k_B T_c = \frac{2J}{\ln(1+\sqrt{2})} = \frac{\epsilon}{2 \ln(1+\sqrt{2})} = 0.567296 \epsilon.$$

Mean-field theory gives $m = \tanh[\beta(qJm + H)]$, hence $k_B T_c^{\text{MF}} = qJ = \epsilon$. It overestimates T_c because it replaces the fluctuating local environment by its mean and neglects spatial correlations.

For a trial flip, put $\delta = 1 - 2c_i = \Delta N$ and let n_i be the number of occupied neighbours. Then

$$\Delta E = -\epsilon \delta n_i, \quad \Delta(E - \mu N) = -\delta(\epsilon n_i + \mu).$$

These are respectively `delta_E`, `delta_N` and `delta_K` in `local_grand_energy_change`.

Validation

After 2000 equilibration sweeps, 12,000 measurements separated by five sweeps gave:

μ	$\langle N \rangle$, simulation	exact	$\text{var}(N)$, simulation	exact
-1	107.522(85)	107.577	78.83(99)	78.64
0	200.056(91)	200.000	99.43(1.28)	100.00
1	292.360(80)	292.423	77.49(91)	78.64

Uncertainties are standard errors from block means (or block variances), using 20-measurement blocks; the measured integrated autocorrelation times were only 0.50–0.52 measurement intervals. The stationary trace and full $P(N)$ at $\mu = -1$ agree with the binomial law (Fig. 1).

At $T = 0.8$, $\langle \rho \rangle_{-2.4} = 0.095389(232)$, while $1 - \langle \rho \rangle_{-1.6} = 0.095104(213)$. Their difference, 0.000285(315), is below one standard error and confirms particle-hole symmetry.

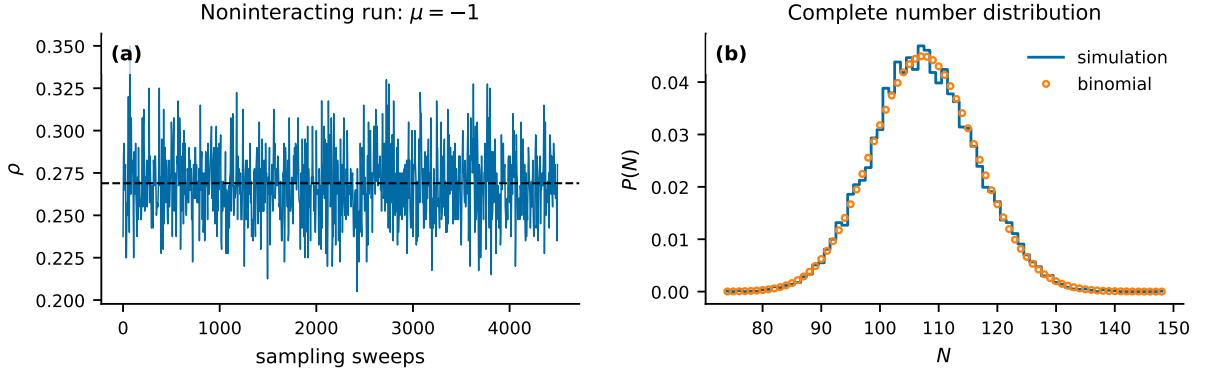


Figure 1: **Validation.** (a) Representative noninteracting density trace after equilibration. (b) The complete measured distribution at $\mu = -1$ and the exact binomial result.

Equation of state and density response

The two equations of state are antisymmetric about $(\mu_{\text{coex}}, \rho) = (-2, 1/2)$ (Fig. 2a). Paired densities satisfy $\rho(-2 + \Delta\mu) + \rho(-2 - \Delta\mu) = 1$ within uncertainty: the largest standardised deviation is 2.0, consistent with the expected scatter among several comparisons. The response becomes much steeper on approaching T_c .

At coexistence the three response estimates were:

T/T_c	finite difference	$\beta V \text{var}(\rho)$	$\beta S(0)$
1.5	1.994(55), $h = 0.01$	2.002(30)	2.002(30)
1.15	13.06(52), $h = 0.005$	13.44(37)	13.44(37)

At $1.5T_c$, a central difference is stable for $h = 0.01$ – 0.02 and agrees with the fluctuation result. At $1.15T_c$, the estimates for $h = 0.05, 0.02, 0.01, 0.005$ are 6.43, 10.25, 11.77, 13.06. Thus a broad equation-of-state grid still underestimates the derivative, but the finest estimate agrees with the fluctuation result within the combined uncertainty. The variance and $S(0)$ estimates coincide because, with the program's normalization, $S(0) = \text{var}(N)/V = V \text{var}(\rho)$; this is an exact normalization check rather than an independent measurement.

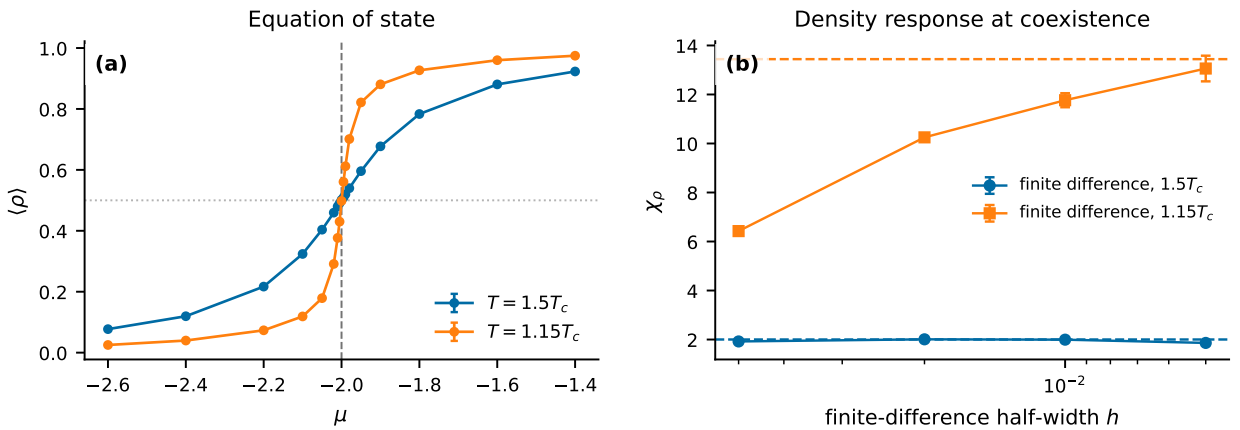


Figure 2: **Equation of state and response.** (a) Density against chemical potential; error bars are mostly smaller than the symbols. (b) Convergence of the symmetric finite-difference derivative as its half-width h is reduced. Dashed horizontal lines are the fluctuation estimates.

Spatial correlations

The configurations develop progressively larger domains as T_c is approached (Fig. 3a–d), accompanied by a pronounced enhancement of $S(k)$ at small k . The returned two-dimensional connected structure factor was averaged over exact square-lattice shells,

$$k = \frac{2\pi}{L} \sqrt{n_x^2 + n_y^2}.$$

Six independent runs were averaged at each temperature. Fits to $S(k) = A/[1 + (k\xi)^2]$ used the first six shells, $0 \leq k \leq 0.7405$; independent-run bootstrap errors and the range obtained by changing $k_{\max}$ are:

T/T_c	ξ	bootstrap s.e.	range for $k_{\max} = 0.55\text{--}0.95$
1.50	1.459	0.010	1.440–1.461
1.25	2.606	0.012	2.574–2.612
1.10	5.801	0.073	5.736–5.824
1.05	9.968	0.058	9.792–10.108

The final point has ξ comparable with $L/2$, and its reduced fit statistic rises to 3.8, signalling finite-size and non-Ornstein–Zernike corrections. A log–log fit gives an effective $\nu_{\text{eff}} = 0.84(1)$; changing $k_{\max}$ moves it only between 0.838 and 0.845. The statistical precision is therefore misleading: finite-size effects, corrections to scaling and the short temperature interval dominate. The growth is appreciably stronger than the mean-field $t^{-1/2}$ prediction and is qualitatively closer to the Ising t^{-1} form, but these data do not justify a precise exponent.

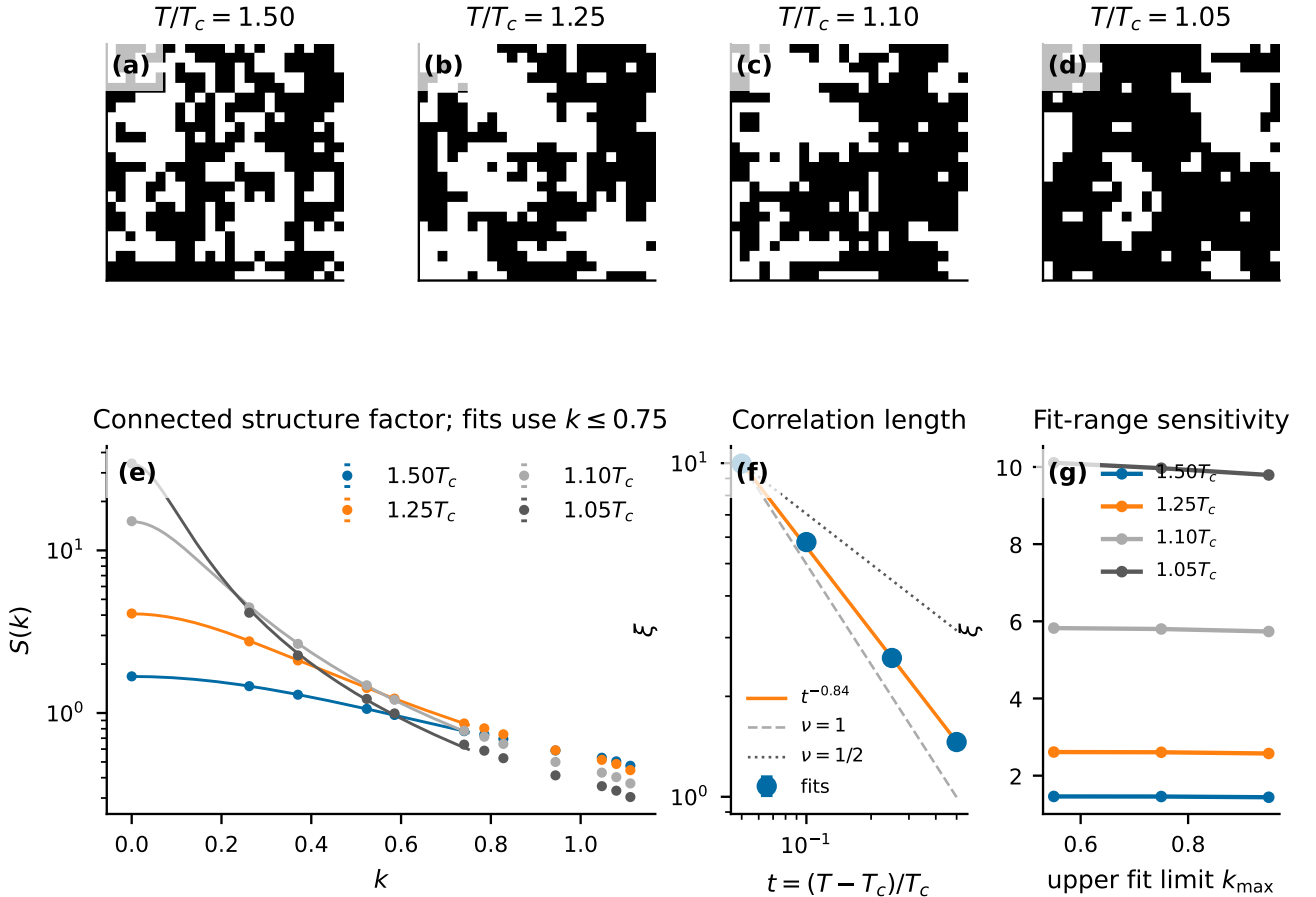


Figure 3: **Spatial correlations.** (a–d) Representative equilibrated configurations. (e) Radially averaged connected structure factors and the fitted low- k forms. (f) Extracted correlation lengths with exact and mean-field guide slopes. (g) Sensitivity to the upper fit limit.

Density distributions and finite-size effects

At $1.2T_c$, $P_L(\rho)$ is a single broad peak centred at $1/2$; the total-variation distance from its reflected form is 0.045. At T_c , the distribution is broad and double peaked. A 160,000-measurement run gives $\langle \rho \rangle = 0.460$ and a reflection distance 0.114: the peak shapes agree well, but slow order-parameter reversals leave imperfect relative sector weights. Below T_c , the two peaks become narrow and lie near $\rho \simeq 0.05$ and 0.95 . The low-start run changed sector three times, whereas the high-start run did not. Residence times are therefore not an equilibrium measure of the weights. Figure 4f uses the exact particle-hole symmetry to reflect the well-sampled high-density sector and assign the two peaks equal weight.

An equilibrium average $\langle \rho \rangle = 1/2$ below T_c would consequently describe an equal mixture of low- and high-density states, not typical homogeneous configurations at density $1/2$. Individual configurations reside overwhelmingly in one phase; the mean lies between the peaks because it averages over phase identity.

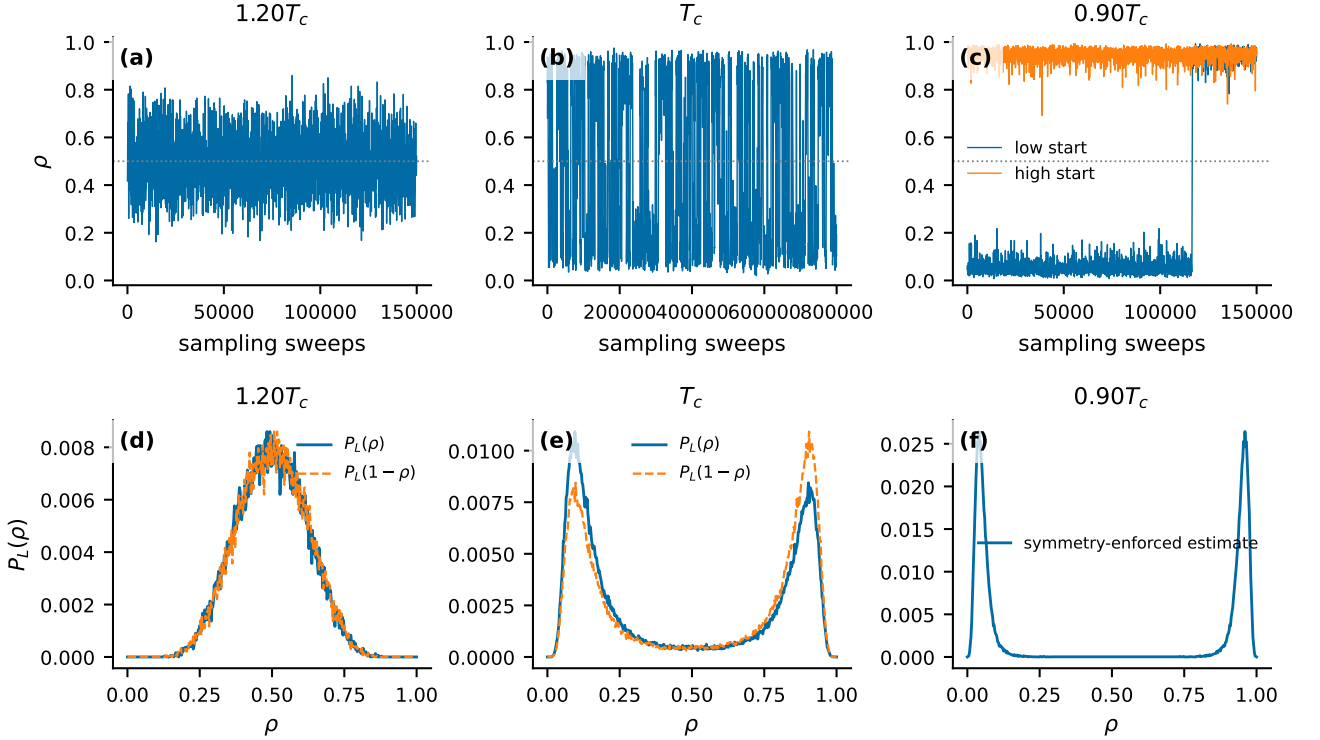


Figure 4: **Evolution of the density distribution.** (a–c) Density histories at coexistence. The subcritical low-start run makes one long excursion into the high-density sector; the high-start run remains trapped. (d,e) Measured distributions and their particle-hole reflections. (f) Subcritical two-peak estimate constructed with the symmetry-justified 1:1 weights.

At criticality, the peaks of $P_L(\rho)$ move towards $1/2$ and the distribution narrows as L increases (Fig. 5a). Two independent 40,000-measurement runs for each size give:

L	U_4	block-bootstrap s.e.
12	0.61064	0.00114
16	0.61131	0.00144
20	0.61110	0.00171
24	0.61110	0.00218

All values agree with $U_4^* = 0.61069$; any finite-size drift is smaller than the present sampling uncertainty. Since U_4 measures dimensionless distribution shape, its approximate independence of L is evidence of scale invariance. In real space this means that critical configurations have no characteristic domain size: density fluctuations occur over a broad range of scales, up to the system size. The uncertainty grows with L because the dynamics slows. For the extended $L = 24$ diagnostic run, $\tau_m = 463$ measurement intervals $= 2.31 \times 10^3$ sweeps, whereas $\tau_{m^2} = 17.7$ intervals; even moments such as U_4 therefore converge much sooner than the relative sign-sector weights. Both sectors were sampled (1355 crossings); first- and second-half distributions differ by only 0.022 in total variation and have means 0.440 and 0.449. The run is adequate for the even moments and distribution shape, but not for a high-precision estimate of the odd quantity $\langle m \rangle = 0$.

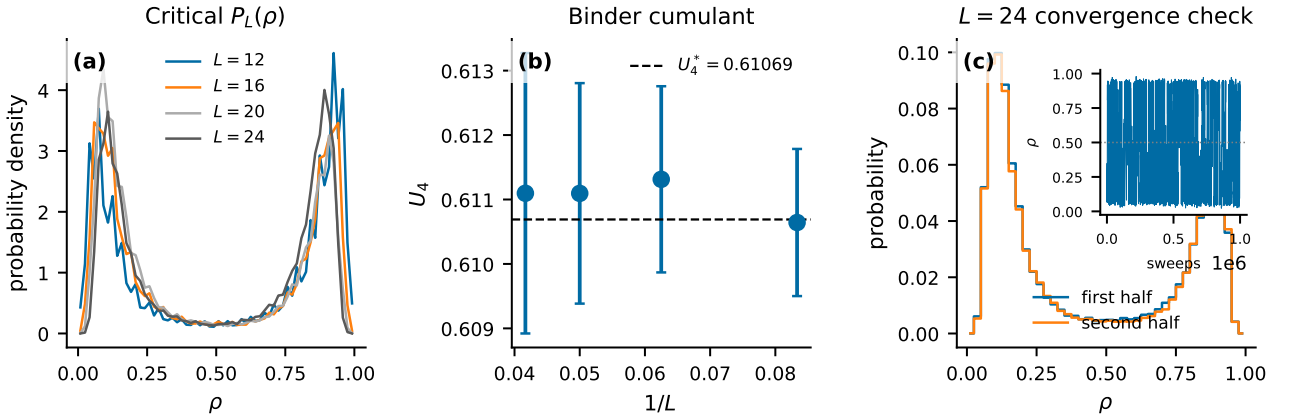


Figure 5: **Critical finite-size behaviour.** (a) Order-parameter distributions. (b) Binder cumulants and the periodic-square reference. (c) First/second-half comparison for the extended $L = 24$ run; inset: its density history.

Production-run record. All seeds are $s = (2468135 + 104729r) \bmod (2^{32} - 1)$, where r is the run ID. The detailed individual record is supplied in `production_run_table.csv`; the complete grouped record used above is:

Purpose; run IDs	L, T, μ, ϵ ; initial density	equilibration; interval; measurements	seed range; acceptance
ideal validation; 1–3	$20, 1, \{-1, 0, 1\}, 0$; exact p	2000; 5; 12000	2572864–2782322; 0.538–1.000
particle-hole; 4–5	$20, 0.8, \{-2.4, -1.6\}, 1$; 0.15/0.85	5000; 5; 12000	2887051–2991780; 0.184
equation of state; 10–31	$24, \{1.5, 1.15\}T_c, -2 +$ $\Delta\mu, 1$; 0.15/0.5/0.85	6000; 5; 12000	3515425–5714734; 0.050–0.533
fine response; 40–51	as above, $\Delta\mu = \pm.02, \pm.01, \pm.005$	6000–8000; 5; 12000–20000	6657295–7809314; 0.324–0.532
correlations; 100–123	$24, \{1.50, 1.25, 1.10, 1.05\}T_c, -2$; 0.5; six replicas	8000; 5; 10000	12941035–15349802; 0.262–0.534
distribution evolution; 200,202–203	$20, \{1.2, 0.9\}T_c, -2, 1$; 0.5/0.1/0.9	6000–10000; 5; 30000	23413935–23728122; 0.090–0.394
critical sizes; 300–307	$L =$ 12, 16, 20, 24, $T_c, -2, 1$; 0.5; two replicas	12000; 5; 40000	33886835–34619938; 0.181–0.191
extended critical; 320–321	$L = 20, 24, T_c, -2, 1$; 0.5	12000; 5; 160000/200000	35981415–36086144; 0.189–0.190

Errors for ordinary means and variances were obtained from blocks at least ten estimated autocorrelation times long. Correlation-length errors resample the six independent runs; U_4 errors use a blocked bootstrap of the two required independent trajectories. Raw trajectories were retained for every entry.

Marking-template note: this model solution was generated for use in marking and therefore does not reproduce the student’s required signed AI-use declaration.